Statistical approach to turbulent dispersal of aerosols for accurate prediction of concentration and associated uncertainties

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In the context of turbulent dispersal of aerosol pollutants from a source within a ventilated indoor space, the present work addresses the importance of going beyond accurate prediction of the mean ensemble-averaged exposure, by evaluating the expected level of variability in individual realizations. This uncertainty quantification requires a statistical description of the inherent stochastic turbulent dispersal process and the inhomogeneous nature of the indoor flow. We leverage large datasets from turbulence-resolving Euler-Lagrange simulations of aerosol dispersal in varying indoor geometries. The datasets provide time-resolved concentrations of pollutants emitted by a source located anywhere in a room and reaching a sink (susceptible individual) that may be located anywhere else in the same room. The statistical information is used to accurately predict the mean pollutant concentration as a function of source-sink separation distance and associated variability in two different experimental measurements to demonstrate the importance of estimating the uncertainty associated with the predicted average concentration. The average pollutant concentration results are compared with the predictions of three other commonly used models (well-mixed model, near-field/far-field model, and eddy diffusivity model).

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I. INTRODUCTION

Understanding how pollutants, aerosols, and virus-laden nuclei [1–6] spread within indoor spaces is of fundamental importance. Such understanding when translated to accurate prediction capability has the potential to improve quality of life, reduce occupational hazards, and save human lives. Consider the case where pollutants originate from a source within a room or are introduced through an inlet vent located along the room's boundary. The quest is to accurately predict the cumulative exposure of a receiving host located elsewhere in the room over a period of time. The predicted level of cumulative exposure can then be used to make important decisions, such as determining safe exposure time, optimal seating arrangements, and ideal ventilation operating conditions.

Despite progress made over the past few decades, accurate prediction of cumulative exposure remains a formidable challenge for the following reasons.

(i) The amount of pollutants reaching a receiving host (henceforth referred to as the sink) depends on advective transport by the mean flow and dispersive transport due to turbulence, which are dictated by the details of natural or forced ventilation [7–9]. In the absence of significant mean

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flow effects, the problem reduces to two-point isotropic turbulent dispersion. However, ventilation-induced mean flow is often strong enough that its effects cannot be ignored. The directionality of mean airflow, especially when the air current is strong, could significantly alter the risk of exposure [10].

(ii) Both the mean flow and turbulence within the indoor space are inhomogeneous. As a result, cumulative exposure at the sink depends not only on its distance from the source but also on where the source and sink are located within the room. In other words, cumulative exposure is an inhomogeneous, anisotropic two-point statistical information that depends on factors such as room size/shape, location of air inlets/outlets, and the mean flow pattern within the room [9,11–13].

(iii) Cumulative exposure is a stochastic quantity with large variability. *It is not sufficient to predict only the expected value of exposure. Standard deviation about the expectation, and perhaps higher order moments if needed, must also be estimated for proper risk assessment*. The stochastic nature of exposure arises from two factors. First, given a separation distance r, there are infinite possible source-sink locations at that separation. Due to the inhomogeneous nature of the indoor flow, there will be substantial variation in cumulative exposure between these different source-sink arrangements. Second, even for a particular source-sink arrangement, exposure remains a stochastic quantity due to randomness in the injection process and turbulent transport.

(iv) Factors such as the extent of exhaust air recycled back into the room and the effectiveness of air filtration for pollutants play a crucial role and must be considered in the prediction model. Ultimately, the focus is on the risk, that is, on the likelihood of the receiving host being adversely affected. This estimation additionally depends on factors such as the emission rate of pollutants by the source, use of protective devices such as a mask, level of exposure that is deemed consequential, etc.

The simplest approach to predicting cumulative exposure has been the classical well-mixed room model (WMM) [14,15]. It assumes instantaneous mixing of pollutants throughout the room and as such neglects effects of proximity between the source and the sink. With this simplification, the WMM becomes independent of the room configuration, ventilation details, and the precise locations of the source and the sink. The clean air exchange rate, expressed in air changes per hour (ACH), becomes the parameter of importance, and recycling and air filtration can be accounted for with the introduction of an effective ACH [16]. The main limitation of the WMM is its inability to account for the source-sink separation distance or to provide an estimation of the uncertainty associated with the predicted cumulative exposure [17].

Near-field/far-field models, zonal models, Markov chain models, and eddy diffusivity models have been developed to address the first deficiency of the WMM [18–24]. In the near-field/far-field model (NFM) [19,25], a box-within-a-box approach is used with the assumption that the air and pollutants are well mixed in each of the boxes. The eddy-diffusivity model (EDM) [20,26] has been a popular choice in predicting exposure to toxic airborne chemicals in indoor spaces [27,28]. EDM uses the analytical solution of the 3D diffusion equation in an unbounded domain with the effect of the room's boundaries taken into account through the method of reflection [29–31].

While WMM considers the pollutant concentration at the sink to be a single-point homogenized statistic, both NFM and EDM approach the concentration as a two-point statistic. However, by considering the pollutant concentration to be a function of only the source-sink separation distance, homogeneity and isotropy are implicitly assumed. In EDM, inhomogeneity due to the bounding walls is approximately accounted for through the method of reflection [32].

The key contribution of the present work is the two-point statistical description of pollutant dispersal, without enforcing homogeneity or isotropy. Using this general description, we obtain expressions for the mean and standard deviations of concentration and cumulative exposure of pollutants due to a continuously emitting source. In this work, we make use of detailed information obtained from large eddy simulations (LES) of flow within indoor spaces to build a higher-order statistical model of turbulent dispersion [16,33]. We avoid assumptions of homogeneity and isotropy by employing the *statistical overloading* technique, which will be expanded upon later in the manuscript.

The mean concentration computed in the LES is then compared with the corresponding results of NFM and EDM. The standard deviation is used to estimate prediction bias, as in the case of the sink being directly downstream (or upstream) of the source, and obtain uncertainties associated with the homogenized predictions. We evaluate the performance of the aforementioned models by comparing against experimental results and demonstrate the importance of going beyond the mean to predict the expected levels of standard deviation. The importance of higher-order nonnormal statistics is measured in terms of skewness and flatness.

The rest of the paper is organized as follows. First, we present the inhomogeneous statistical framework. With this framework, the WMM, NFM, and EDM are examined using LES results of flow and pollutant dispersal in indoor spaces [33,34]. We then evaluate the models against two experimental studies analyzing aerosol dispersal for continuous release from a point source [29,30].

II. STATISTICAL DESCRIPTION

Given the location of a source ($\mathbf{x}_{so}$) and a sink ($\mathbf{x}_{si}$) within an indoor space of volume $\mathcal{V}$, the pollutant concentration at the sink, n, remains a stochastic quantity. At a minimum, it is a function of seven variables: $\mathbf{x}_{so}$, $\mathbf{x}_{si}$, and time t elapsed since the initiation of pollutant ejection at the source. In addition, n will depend on other details such as room size and shape, effective ACH, location of air inlets and outlets, etc. Another parameter of importance is the radius R of the aerosol, which can be taken to be approaching zero in the case of molecular pollutants. With increasing values of R, gravitational removal by settling plays a role. We collectively denote R and all other parameters by the set ψ.

Pollutant concentration $n(t, \mathbf{x}_{so}, \mathbf{x}_{si}; \psi)$ is a stochastic quantity, due to the turbulent nature of both the emission and the dispersion processes. In other words, given the values of t, $\mathbf{x}_{so}$, $\mathbf{x}_{si}$, and ψ, the concentration n at the sink will vary from one realization (one experimental measurement) to the next. The ensemble-averaged concentration is denoted as $\langle n\rangle(t, \mathbf{x}_{so}, \mathbf{x}_{si}; \psi)$. The above description recognizes the inhomogeneous nature of turbulence within the indoor space whereby $\langle n\rangle$ is a function of the two vectors $\mathbf{x}_{so}$ and $\mathbf{x}_{si}$. This dependence emphasizes the important fact that the ensemble-averaged concentration varies depending on the location of the source and sink within the room.

The pollutant concentration statistics can be simplified by *homogenizing* the two-point statistics as

$$\langle\langle n\rangle\rangle(t, r; \psi) = \frac{1}{I_{ho}} \iint_{|\mathbf{x}_{so}-\mathbf{x}_{si}|=r} \langle n\rangle \, d\mathbf{x}_{so} \, d\mathbf{x}_{si}, \tag{1}$$

where the double integration is over all possible source and sink locations, such that the distance between the two is r, and the normalization factor $I_{ho} = \iint_{|\mathbf{x}_{so}-\mathbf{x}_{si}|=r} d\mathbf{x}_{so} \, d\mathbf{x}_{si}$. The integration spans all possible source-sink positions and orientations and therefore involves the isotropy approximation. The result is that $\langle\langle n\rangle\rangle$ is now only a function of the two variables t and r (apart from ψ) and thus far more amenable to modeling than $\langle n\rangle$. In fact, both NFM and EDM can be considered as approximating $\langle\langle n\rangle\rangle$ and thus can be applied for any source and sink combination within the indoor space with only the separation distance r as the variable [35].

A. Characterization of statistical variation

Two sources of variations can be distinguished from a turbulent flow perspective. The first is the statistical variation of an individual realization from the ensemble average. This variation is given by the distribution of n that varies about the ensemble average $\langle n\rangle$. This distribution accounts for the random emission process at the source and the turbulent nature of the pollutant-laden air parcels reaching the sink. A good measure of this variability is the standard deviation

$$\sigma_1(t, \mathbf{x}_{so}, \mathbf{x}_{si}, \psi) = \langle n'^2\rangle^{1/2} = \langle (n - \langle n\rangle)^2\rangle^{1/2}. \tag{2}$$

This standard deviation is an inhomogeneous function of the source and sink locations. Upon homogenization with a spatial average, we obtain

$$\langle\langle\sigma\rangle\rangle_1^2(t,r;\psi) = \frac{1}{I_{\text{ho}}}\iint_{|\mathbf{x}_{\text{so}}-\mathbf{x}_{\text{si}}|=r} \sigma_1^2\, d\mathbf{x}_{\text{so}}\, d\mathbf{x}_{\text{si}}, \tag{3}$$

which is now independent of the specific source and sink locations. Again, for consistency, we have used the double angle bracket to denote the average over both the ensemble as well as all possible source-sink combinations.

The second variation arises from the fact that for a given separation distance r, $\langle n\rangle$ calculated for different combinations of $\mathbf{x}_{\text{so}}, \mathbf{x}_{\text{si}}$ substantially deviates from the homogenized value $\langle\langle n\rangle\rangle$. The standard deviation of this spatial variation may be expressed as

$$\langle\langle\sigma\rangle\rangle_2^2(t,r;\psi) = \frac{1}{I_{\text{ho}}}\iint_{|\mathbf{x}_{\text{so}}-\mathbf{x}_{\text{si}}|=r} (\langle n - \langle\langle n\rangle\rangle)^2\, d\mathbf{x}_{\text{so}}\, d\mathbf{x}_{\text{si}}. \tag{4}$$

This variation is the added uncertainty in the concentration prediction when a homogenized model is used.

We are often interested in predicting the pollutant concentration at a receiving host (i.e., at the sink) in a particular experimental realization, where the source and sink are separated by a distance r. The pollutant concentration measured in an individual realization $n(t,\mathbf{x}_{\text{so}},\mathbf{x}_{\text{si}};\psi)$ will deviate from the homogenized concentration $\langle\langle n\rangle\rangle(t,r;\psi)$ due to both turbulent fluctuations and homogenization over all source-sink combinations. The standard deviation of this total statistical variation is expressed as

$$\langle\langle\sigma\rangle\rangle^2(t,r;\psi) = \frac{1}{I_{\text{ho}}}\iint_{|\mathbf{x}_{\text{so}}-\mathbf{x}_{\text{si}}|=r} \langle n - \langle\langle n\rangle\rangle\rangle^2\, d\mathbf{x}_{\text{so}}\, d\mathbf{x}_{\text{si}}. \tag{5}$$

If we assume the two variations to be statistically independent, then

$$\langle\langle\sigma\rangle\rangle^2 = \langle\langle\sigma\rangle\rangle_1^2 + \langle\langle\sigma\rangle\rangle_2^2. \tag{6}$$

The key objective of this work is to present estimates of both $\langle\langle\sigma\rangle\rangle_1$ and $\langle\langle\sigma\rangle\rangle_2$ using turbulence theory and LES results and thereby obtain an accurate prediction of the standard deviation $\langle\langle\sigma\rangle\rangle$ in addition to the mean $\langle\langle n\rangle\rangle$.

B. Mean and variance of concentration

Consider the problem of turbulent dispersion of a scalar field from a point source of continuous release in isotropic turbulence. The objective is to obtain solutions of both the mean concentration $\langle n\rangle$ and the associated standard deviation $\langle n'^2\rangle^{1/2}$ at a point r away from the source. In the isotropic limit, the double average is not necessary since $\langle n\rangle$ is only a function of r.

The pollutant concentration statistics is spherically symmetric and the mean scalar concentration equation can be expressed as

$$\frac{\partial\langle n\rangle}{\partial t} \approx \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 K\frac{\partial\langle n\rangle}{\partial r}\right) + S_n, \tag{7}$$

where the radial Reynolds flux of pollutant due to turbulence is modeled as $K\partial\langle n\rangle/\partial r$, and the effects of molecular diffusivity are ignored. K is the isotropic turbulent diffusivity of the scalar, which is typically much larger than the molecular diffusivity. In general turbulent diffusivity is a tensorial quantity, which in the present spherically symmetric context becomes a scalar quantity directed only in the radial direction. Furthermore, diffusivity will in general depend on the radial position and must be represented as $K(r)$. Here we will approximate it to be a scalar constant, which is appropriate for larger separation distance.

In Eq. (7), S_n represents the source of pollutants located at $r = 0$ such that its volume integral yields $q_{\text{so}} = \int S_n\, dV$. The turbulent diffusivity K is typically taken to be a constant in time and

space. For a spherically symmetric and continuous point source, Eq. (7) can be solved, assuming K to be a constant as in Refs. [26,32], to obtain

$$\langle n\rangle_{\rm th}(t,r) = \frac{q_{\rm so}}{(4\pi Kr)}\ \operatorname{erfc}\left(\frac{r}{2\sqrt{Kt}}\right), \tag{8}$$

where $\langle n\rangle_{\rm th}(t,r)$ represents the mean concentration at time t at a radial distance r. The subscript "th" has been added to denote the theoretical nature of the prediction.

Standard Deviation of Concentration. The pollutants released at the source reach the sink as randomly carried by turbulent eddies. The pollutant concentration at the sink is thus a fluctuating quantity, with its mean specified in Eq. (7), and whose statistical variance is given by [36]

$$\frac{\partial\langle n'^2\rangle}{\partial t} \approx \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2K\frac{\partial\langle n'^2\rangle}{\partial r}\right) + 2K\left(\frac{\partial\langle n\rangle}{\partial r}\right)^2 - C_n\frac{\epsilon}{k}\langle n'^2\rangle + S_{n'^2}. \tag{9}$$

The first term on the right-hand side of Eq. (9) accounts for the Reynolds flux of variance. This term employs the approximation $-\langle u_r' n'^2\rangle = K\partial\langle n'^2\rangle/\partial r$, whereby effects of molecular diffusivity are ignored. The second term is the production term that goes as the square of the mean concentration gradient. This term can be explicitly calculated from Eq. (8). The third term is a simple model for dissipation of $\langle n'^2\rangle$, where C_n is a constant, and ϵ and k are the dissipation rate and fluctuation kinetic energy of the isotropic turbulence, respectively. In essence, $C_n\epsilon/k$ must be determined, along with K, based on the level of ambient turbulence. Finally, $S_{n'^2}$ represents the variation introduced by a fluctuating source.

Equation (9) can be solved to obtain a measure of concentration fluctuation. Singularity at $r = 0$ arising from the production term is avoided by seeking the solution for $r \geqslant r_0$, with the following boundary condition: $\langle n'^2\rangle(t,r_0) = \sigma_1^{*2}(t,r_0)\,\langle n\rangle^2(t,r_0)$, where $\sigma_1^*(t,r_0)$ is the normalized standard deviation of concentration fluctuation induced by both the source and the near-field turbulence around the source ("*" indicates a normalized quantity).

The analytical steady-state solution for $\langle n'^2\rangle_{\rm ss}$ in the limit $t\to\infty$ for a steady continuous source is presented below. Here again we approximate the diffusivity $K(r)$ to be constant in obtaining the solution. The analytical solution outside a region of radius r_0 measured from the source can be expressed as

$$\begin{aligned}\langle n'^2\rangle_{\rm ss} = {}& \frac{\beta\sqrt{\tilde\epsilon}}{4r}\exp(-r\sqrt{\tilde\epsilon})[\mathrm{Ei}(r\sqrt{\tilde\epsilon}) - \mathrm{Ei}(r_0\sqrt{\tilde\epsilon})] - \frac{\beta\sqrt{\tilde\epsilon}}{4r}\exp(-r\sqrt{\tilde\epsilon})\\ &\times\left[\mathrm{Ei}(-r\sqrt{\tilde\epsilon})\exp(r^2\tilde\epsilon) - \mathrm{Ei}(-r_0\sqrt{\tilde\epsilon})\exp\left(r_0^2\tilde\epsilon\right)\right]\\ &+ \frac{\beta}{2rr_0}\exp(-\sqrt{\tilde\epsilon}(r-r_0))[\phi_0+1] - \frac{\beta}{2r^2},\end{aligned} \tag{10}$$

where Ei is the exponential integral function and the subscript *ss* indicates steady state. In the above, the inner boundary condition near the origin is given by $\phi_0 = \sigma_1^{*2}(t\to\infty, r_0)$. The parameters $\beta = q_{\rm so}^2/(8\pi^2K^2)$ is the scaled source and $\sqrt{\tilde\epsilon} = \sqrt{C_n\epsilon/(kK)}$ is the inverse turbulence length scale.

Figure 1 illustrates the behavior of mean and standard deviation of pollutant concentration obtained from LES of a continuous point source located at the center of a large room. The response to continuous release was approximated by concatenating together responses to consecutive discrete delta function emissions of 60 s periods. Panel A shows pollutant concentration normalized by $q_{\rm so}$ from 8 different realizations of turbulent dispersal pertaining to two source-sink separation distances of $r = 0.8$ and 2 m. Here t represents the elapsed time from the initiation of pollutant ejection at the source. In all the realizations, the time evolution of concentration at the sink is nonmonotonic and the instantaneous fluctuations are dictated by the amount of pollutant within the pockets of turbulent flow arriving at the sink. The level of fluctuation decreases with increasing source-sink separation distance, r. Despite the turbulent fluctuations, all the realizations show a general trend

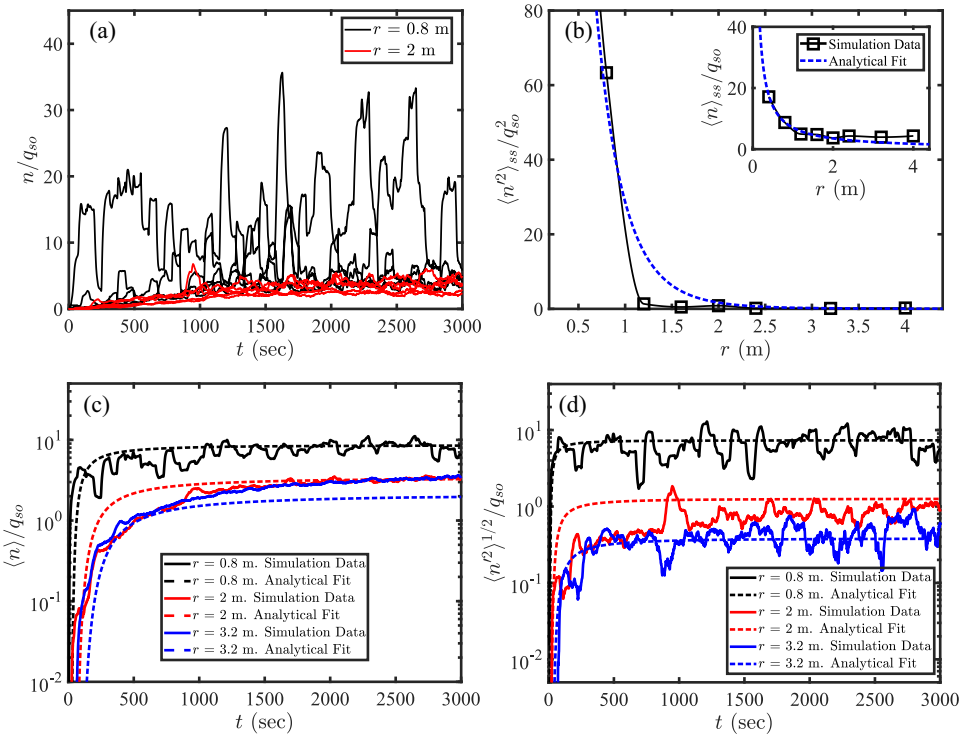


FIG. 1. (a) Time evolution of normalized concentration for a continuously emitting source located at the center of the canonical room [see Figs. 2(a)–2(c)]. The lines represent five of the eight realizations obtained from simulation data for $r = 0.8$ and 2 m. (b) Steady-state values of the normalized variance evaluated from the eight realizations plotted as a function of source-sink separation distance, r along with the theoretical prediction given in Eq. (10). Also plotted in the inset are the steady-state values of the normalized mean concentration obtained from the simulation along with the theoretical prediction given in Eq. (8). (c) Time evolution of normalized mean concentration and (d) normalized standard deviation plotted for $r = 0.8$, 2, and 3.2 m (solid lines). Also plotted in panels (c) and (d) are the corresponding temporal evolutions predicted by the theory (dashed lines).

of concentration increasing over time toward a steady-state value. However, the level of variation between the different realizations is substantial and persists for a long time.

Figure 1(b) shows the steady-state values of the normalized mean $\langle n \rangle_{ss}/q_{so}$ (in the inset) and the steady-state value of normalized variance $\langle n'^2 \rangle_{ss}/q_{so}^2$. The steady-state value is observed to decrease with increasing r in reasonable agreement with the theoretical prediction given in Eq. (8). Also plotted in the figure is the theoretical prediction of the steady-state value of the variance given in Eq. (10) evaluated with the following best fit parameters: $K = 0.0113$ m^2/s, $\tilde{\epsilon} = C_n\epsilon/kK = 8.124$ m^{-2}, $r_0 = 0.186$ m, and $\phi_0 = \sigma_1^{*2}(t \to \infty, r_0) = 0.173$. Again reasonable agreement between the theory and the simulation results is observed.

Figures 1(c) and 1(d) show time evolution of the ensemble-averaged mean $\langle n \rangle/q_{so}$, and the time evolution of the standard deviation $\langle n'^2 \rangle^{1/2}/q_{so}$. The temporal evolution of the ensemble-averaged mean concentration is similar across the three values of r. Although fluctuations persist due to lack of adequate number of samples for converged evaluation of the true ensemble average, we can observe the mean to approach the steady-state value on a time scale of the order of an hour, which is consistent with the expected inverse time scale of 3600/ACH, for an ACH of 2.5. The timescale of standard deviation [Fig. 1(d)] is however substantially shorter. The steady-state value of both the ensemble-average and the standard deviation decreases with increasing r.

Also plotted in Figs. 1(c) and 1(d) are the theoretical predictions of the time evolution. For the ensemble averaged concentration the time-dependence is given in Eq. (8). For the variance, rather than seeking a time-dependent analytical solution to Eq. (9), we propose the following temporal evolution based on the LES results:

$$\langle n'^2\rangle_{\rm th}^{1/2}(t,r) = \langle n'^2\rangle_{\rm ss}^{1/2}(r)\ \ {\rm erfc}\left(\frac{r r_0\sqrt{\tilde{\epsilon}}}{2\sqrt{Kt}}\right). \tag{11}$$

The values of $\tilde{\epsilon}$ and r_0 are those obtained by fitting the steady-state values of the variance. From the reasonable agreement seen in Fig. 1(d), we observe the diffusivity of standard deviation to go as $K/\tilde{\epsilon}\, r_0^2$. This explains the substantially shorter duration to reach steady state for the standard deviation as seen in Fig. 1(d).

III. SINGLE- AND TWO-POINT MODELS

In this section, we discuss four different models that can be used to predict the pollutant concentration at a receiving host. The four models are:

(i) The well-mixed room model: This single-point homogenized model can only predict the room-averaged concentration, denoted as $\langle\langle n\rangle\rangle_{\rm wmm}(t;R)$ and cannot estimate statistical variation.

(ii) The near-field far-field model: This is a two-point homogenized model that accounts for source-sink separation but does not provide estimates of statistical variation. We denote its prediction as $\langle\langle n\rangle\rangle_{\rm nfm}(t,r;R)$.

(iii) The eddy diffusivity model: This is also a two-point model that accounts for the source-sink separation. By including the presence of domain boundaries the model is rendered inhomogeneous and the general prediction becomes $\langle n\rangle_{\rm edm}(t,\mathbf{x}_{\rm so},\mathbf{x}_{\rm si};R)$. This model can be homogenized to obtain $\langle\langle n\rangle\rangle_{\rm edm}(t,r;R)$.

(iv) The present data-driven stochastic model: By construction, this is a two-point inhomogeneous model that accounts for the precise locations of the source and sink. The pollutant concentration is predicted as $\langle n\rangle_{\rm dsm}(t,\mathbf{x}_{\rm so},\mathbf{x}_{\rm si};R)$, which can be homogeneized to obtain $\langle\langle n\rangle\rangle_{\rm dsm}(t,r;R)$. This model provides estimates of statistical variations due to both turbulence and spatial inhomogeneity as $\langle\langle\sigma\rangle\rangle_{1,\rm dsm}$ and $\langle\langle\sigma\rangle\rangle_{2,\rm dsm}$.

Single-point well-mixed model. For a constant continuous source of pollutant release of magnitude $q_{\rm so}$ starting at time $t=0$, the temporal evolution of pollutant concentration according to the WMM can be expressed as [14,15,37,38]

$$\langle\langle n\rangle\rangle_{\rm wmm}(t;R) = \frac{q_{\rm so}}{\lambda(R)\mathscr{V}}(1-e^{-\lambda(R)t}). \tag{12}$$

The well-mixed prediction is independent of r and the removal rate in Eq. (12) is given by $\lambda(R) = {\rm ACH}/3600 + V_s(R)/H + \lambda_{\rm da}$, where $V_s(R)$, H, and $\lambda_{\rm da}$ represent the size-dependent settling velocity of the pollutant aerosol, the room height, and the exponential rate of pollutant deactivation, respectively.

Near-field far-field model. NFM divides the indoor space into two well-mixed boxes with limited exchange of air between them. The near-field is the zone containing the source of emission and the sink where the exposure is being estimated. The far-field is the remainder of the room. The concentration in the near- and far-field regions is obtained through the solution of the coupled mass balance equations. It is assumed that both fields are internally well-mixed with a constant interzonal flow rate, denoted as β. The complete solution is tedious, and frequently it is simplified [25]. For a constant rate of emission $q_{\rm so}$ and pollutant removal rate $\lambda(R)$, the predicted homogeneized pollutant concentration is

$$\langle\langle n\rangle\rangle_{\rm nfm}(t,r;R) = \frac{q_{\rm so}}{\beta}\left(1-\exp\left(-\frac{\beta}{\mathscr{V}_{\rm NF}}t\right)\right) + \langle\langle n\rangle\rangle_{\rm wmm}(t;R). \tag{13}$$

For the NF model, it is recommended [25,39] that the interzonal flow rate β, be calculated from the relation, $\beta = \frac{1}{2}\text{FSA}\, s$, where FSA is the free surface area of the near field volume and s is the random air speed within the room. The median random air speed observed in typical indoor office and home spaces is between 0.05 and 0.1 m/s [40]. Nicas [41] recommends a value of 0.06 m/s for typical indoor settings.

Eddy diffusivity model. The eddy diffusivity model starts with the spherically symmetric mean scalar concentration equation and solution given earlier in Eqs. (7) and (8). These isotropic results are then improved [26,30] by incorporating an exponential decay term to account for the effect of ventilation, and reflected sources to account for the containment effect of the walls. The model, however, does not account for variation introduced by the room-scale mean flow. The homogenized mean concentration $\langle\langle n\rangle\rangle_{\text{edm}}(t, r; R)$, and the standard deviation $\langle\langle \sigma\rangle\rangle_{2,\text{edm}}(t, r; R)$ of the eddy-diffusivity model can be calculated as defined in Eqs. (1) and (4). Implementation details are expanded upon in the Supplemental Material (SM) [42] (including Ref. [43]).

The only closure parameter needed is the turbulent diffusivity K. Experimental observations have shown values of K ranging from 1×10^{-3} m^2/s [26] to 1.9×10^{-1} m^2/s [20,30,44]. However, there is limited data regarding the variation of K across different rooms and ventilation systems. Venkatram & Weil [45] observe the following linear relationship: $K = \mathscr{V}^{2/3}\text{ACH}/3600$. Foat *et al.* [46] obtain a similar relationship, given by $K = 0.824N^{-2/3}\mathscr{V}^{2/3}\text{ACH}/3600$ where N represents the total number of inlet ventilation vents.

Data-driven stochastic model. The stochastic model uses the Lagrangian particle trajectories computed in the LES simulations [33,34]. The geometric setup of the LES is a 10 m × 10 m × 3.2 m room with a four-way cassette ventilation unit located on the ceiling [Fig. 2(c)]. The outlet is centrally located with four symmetrically located inlet ports. The room is statistically overloaded at the start of the simulation with 20 million aerosol particles of 10 different radii (0.1, 0.5, 2.5, 5, 7.5, 10, 12.5, 15, 20, and 25 μm). In this work, we only consider tracer-like pollutant aerosols of radii less than a micron (i.e., in the $R \to 0$ limit) whose gravitational settling is negligible. Each aerosol is tracked individually over the entire duration of the simulation. The room is discretized into 5000 subvolumes [Fig. 2(b)] and from the Lagrangian statistics, the probability of aerosol traveling from any of the 5000 subvolumes to any other subvolume over a time interval t is calculated. Thus, we consider 25 million different source-sink combinations within the room and obtain accurate converged statistics of both the mean concentration $\langle n\rangle(t, \mathbf{x}_{\text{so}}, \mathbf{x}_{\text{si}}; R)$, and standard deviation $\sigma_1(t, \mathbf{x}_{\text{so}}, \mathbf{x}_{\text{si}}; R)$. For each of these source-sink combinations, the response to a discretized delta function release is calculated by following the Lagrangian trajectories of aerosols that were observed at the source over a 60 s period and observing if and when they arrive at the sink. For a continuously emitting source, the concentration of pollutant at the sink at time t is then calculated by integrating the delta function response from 0 to t. Validation of this method against other ways of calculating the sink concentration for a continuously emitting source is presented in the SM [42].

It is important to note that the statistical information gathered from the computed Lagrangian trajectories is voluminous since it depends not only on t, but also on every source-sink combination. These quantities are then ensemble averaged to obtain $\langle n\rangle_{\text{dsm}}$ and $\sigma_{1,\text{dsm}}$. These results are then homogenized as shown in Eqs. (1), (3), (4), and (5) to obtain $\langle\langle n\rangle\rangle_{\text{dsm}}$, $\langle\langle \sigma\rangle\rangle_{1,\text{dsm}}$, $\langle\langle \sigma\rangle\rangle_{2,\text{dsm}}$, and $\langle\langle \sigma\rangle\rangle_{\text{dsm}}$. The homogenized simulation results and the standard deviation are expressed in terms of the WMM and an r-dependent correction function $\langle\langle \gamma\rangle\rangle$ and its standard deviation $\langle\langle \sigma\rangle\rangle_{\gamma}$ as

$$\langle\langle n\rangle\rangle_{\text{dsm}}(t, r; R) = \langle\langle n\rangle\rangle_{\text{wmm}}(t; R)\, \langle\langle \gamma\rangle\rangle(t, r; R). \tag{14}$$

$$\langle\langle \sigma\rangle\rangle_{\text{dsm}}(t, r; R) = \langle\langle n\rangle\rangle_{\text{wmm}}(t; R)\, \langle\langle \sigma\rangle\rangle_{\gamma}(t, r; R). \tag{15}$$

Although here we focus only on the mean and standard deviation, the comprehensive Lagrangian statistics allow for the calculation of higher-order statistics such as skewness and kurtosis, which are presented in the SM [42].

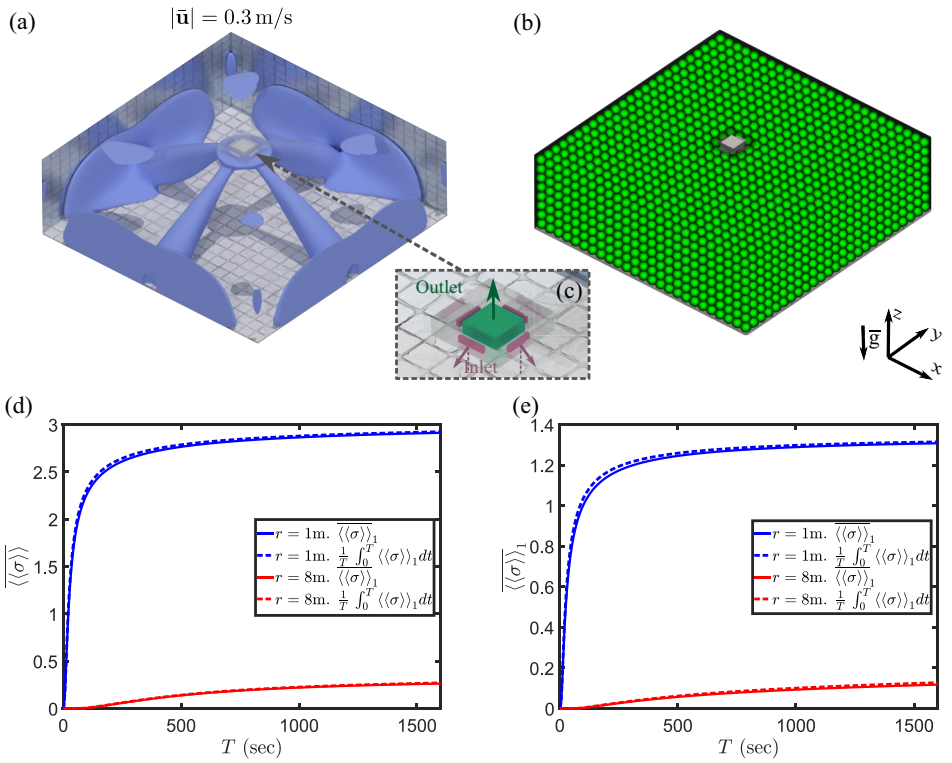


FIG. 2. (a) Isocontours of time-averaged velocity magnitude of 0.3 m/s for an ACH of 5 in a canonical room ($10 \times 10 \times 3.2$ m^3) with an (c) overhead cassette ventilation unit. (b) A depiction of the room being divided into 5000 subvolumes for statistical post-processing. Also plotted are comparisons of (d) $\overline{\langle\langle\sigma\rangle\rangle}$ (solid lines) and (e) $\overline{\langle\langle\sigma\rangle\rangle}_1$ (solid lines) along with their approximated value as per the right-hand side of Eq. (19) (dashed lines) for $r = 1$ and 8 m.

A. Model modification for time-averaged concentration

Often, the quantity of interest is time-averaged concentration at the sink. The time average over a period $0 \leqslant t \leqslant T$ is defined with an overbar as

$$\overline{n}(T, \mathbf{x}_{\mathrm{so}}, \mathbf{x}_{\mathrm{si}}; R) = \frac{1}{T}\int_0^T n(t, \mathbf{x}_{\mathrm{so}}, \mathbf{x}_{\mathrm{si}}; R)\, dt. \tag{16}$$

Here it is assumed that the source and sink enter the room at the same time. Often this is not the case, and the limits of time integration can be modified accordingly. The mean and standard deviation of cumulative exposure will now be defined in terms of instantaneous values. The time average commutes with both ensemble average and conditional integrations over all source-sink combinations and as a result

$$\overline{\langle\langle n\rangle\rangle}(T, r; R) = \frac{1}{T}\int_0^T \langle\langle n\rangle\rangle(t, r; R)\, dt. \tag{17}$$

Similarly, the standard deviation of time-averaged concentration can be defined as

$$(\overline{\langle\langle\sigma\rangle\rangle}(T, r; R))^2 = \frac{1}{I_{\mathrm{ho}}}\iint_{|\mathbf{x}_{\mathrm{so}}-\mathbf{x}_{\mathrm{si}}|=r} \langle(\overline{n} - \overline{\langle\langle n\rangle\rangle})^2\rangle\, d\mathbf{x}_{\mathrm{so}}\, d\mathbf{x}_{\mathrm{si}}. \tag{18}$$

$\overline{\langle\langle\sigma\rangle\rangle}$ cannot be simply expressed as the time integral of the instantaneous standard deviation as was done with the mean, $\overline{\langle\langle n\rangle\rangle}$. From the simulation data, $\overline{\langle\langle\sigma\rangle\rangle}$ was computed directly following (18). We also computed the integral $(1/T)\int_0^T \langle\langle\sigma\rangle\rangle\, dt$. The comparison shown in Figs. 2(d) and 2(e) shows the relation

$$\overline{\langle\langle\sigma\rangle\rangle} \approx (1/T)\int_0^T \langle\langle\sigma\rangle\rangle\, dt \tag{19}$$

to approximately hold. This offers the advantage that neither $\overline{\langle\langle n\rangle\rangle}$ nor $\overline{\langle\langle\sigma\rangle\rangle}$ need to be modeled independently.

IV. COMPARISON OF MODELS

In this section, we evaluate the accuracy of the WMM, EDM, and NFM in predicting pollutant concentration, whereby we use $\langle\langle n\rangle\rangle$ obtained from LES [33] as the ground truth. Since this information was used in the definition of $\langle\langle\gamma\rangle\rangle$, the DSM is deemed exact and offers the basis against which the other three models are compared. The simulation data used corresponds to pollutant concentration of submicron particles at an ACH of 5 with perfect filtration.

The tunable parameters of EDM and NFM, i.e., K and β, respectively, are obtained by fitting the models to the time-averaged LES concentrations for a separation distance of 1 m. For NFM, β is dependent on the volume of the near-field, which varies with separation distance. Therefore, the best-fit value of β for $d = 1$ m is used to obtain the random air speed, s, which is used to evaluate β at all other separation distances.

The best fit for the NFM at $r = 1$ m yields a value of $\beta = 0.735$ m^3/s. This yields a random air speed of $s = 0.117$ m/s, which is in the range observed by Baldwin and Maynard [40]. This is also consistent with the mean velocity magnitude of 0.14 m/s observed in the simulation [47] for ACH $= 5$. The best-fit values of K obtained 0.0455 m^2/s, which is consistent with mean diffusivity that can be estimated with room-averaged dissipation and kinetic energy. The corresponding values of K obtained using the equations provided by Venkatram and Weil [45] and Foat *et al.* [48] for an ACH of 5 are 0.0650 m^2/s and 0.0212 m^2/s, respectively, which are reasonably close to the best fits.

Figure 3 shows the LES data against the predictions of NFM and EDM. Figure 3(a) shows the time-averaged concentration over $T = 1600$ s, normalized by the WMM prediction for $r =$ ranging from 0.4 to 8 m. The remaining panels show the temporal evolution of the concentration normalized by the steady-state well-mixed concentration at (b) $r = 0.4$, (c) 1, and (d) 8 m. The WMM is also included in the plots to illustrate variations in concentration relative to the room-averaged concentration. Note that by definition, as $t \to \infty$, the WMM will approach unity. In this comparison, we have taken the pollutant aerosols to be very small in size with negligible decay rate (i.e., we have set $R \to 0$ and $\lambda_{da} = 0$).

We first describe the observed behavior of pollutant dispersal in the simulations (black lines in Fig. 3). The time-averaged concentration in Fig. 3(a) is observed to be higher (or lower) than the well-mixed prediction at separation distances smaller (or larger) than about 4 m. In Fig. 3(b), for $r = 0.4$ m, the short-time rapid increase is due to advection by the room-scale mean flow, whereas the long-time slower increase is due to turbulent dispersion. Furthermore, as r increases, the initial rapid increase by advection diminishes. In Figs. 3(b) and 3(c), the simulation reveals higher concentrations relative to the WMM prediction, whereas in Fig. 3(d) the trend reverses.

In Fig. 3(a), NFM (green curve) coincides with the LES result at $r = 1$ m. By choosing a different β the green curve can be shifted up or down to be in better agreement at any other r. The relative error is around 56% at $r = 0.4$ m, and the error does not exceed 51% at larger separation distances. The rapid short-time advective increase followed by a long-time slower dispersive increase is well captured by the discontinuous nature of NFM. The overprediction is due to the short-time advective component. NFM is mainly intended to be used in the near-field region of the source with the intent to account for the increase in concentration near the source above the room average. As can be seen

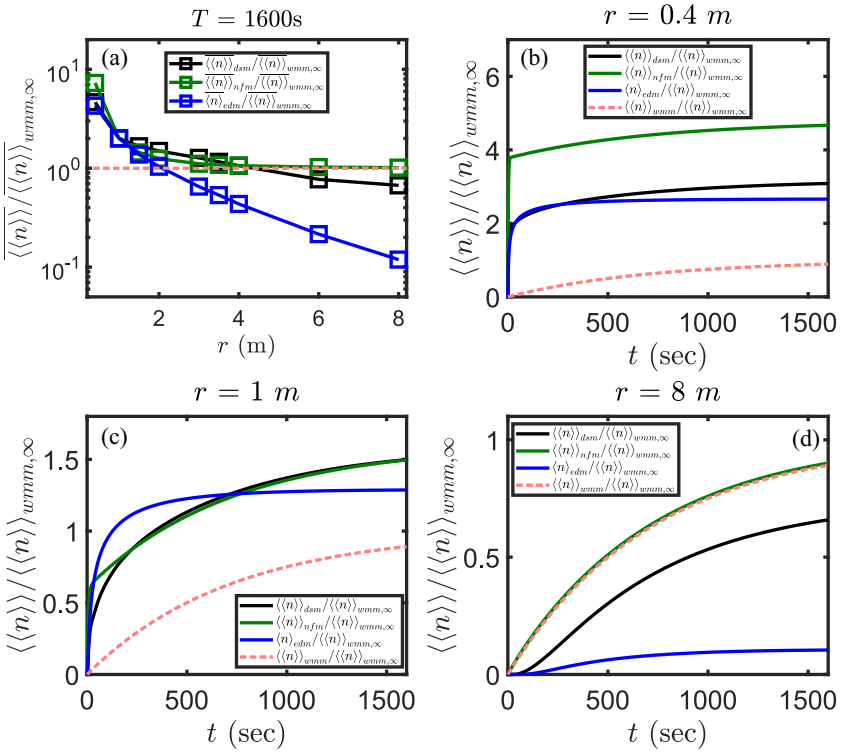


FIG. 3. Comparison of WMM, EDM, and NFM against simulation data (DSM) using the best-fits of the tunable parameters K and β with simulation data at $r = 1$ m. (a) Normalized time-averaged concentration vs separation distance. Normalized concentration vs time at separation distances of (b) 0.4 m, (c) 1 m, and (d) 8 m.

in Eq. (13), the second term on the right-hand side is the same as the room average predicted by the WMM, and the addition of the first term aims to account for the increased near-field contribution. Thus, the NFM prediction is guaranteed to be larger than the WMM. This leads to an overprediction at the larger separation distances, where the actual concentration is expected to be lower than the well-mixed average.

An examination of the EDM model shows that the time-averaged concentration should follow a linear decrease with increasing r at shorter distances, which changes to an exponential decay at larger separation distances. Figure 3(a) is in agreement with this expected behavior. We observe errors of the order of 80% at large separations, which can be seen in the temporal evolutions presented in Figs. 3(b)–3(d). Although the EDM seems to perform reasonably well at shorter separation distances [Fig. 3(b)], it significantly underpredicts the concentration at larger separation distances [Fig. 3(d)] despite accounting for the effect of reflection. While the steady-state concentration predicted by the EDM can be significantly lower at larger separation distances, it captures the nature of the temporal evolution wherein we see a period of time where the concentration nearly remains zero before starting to increase.

A. Standard deviation of concentration

Here we focus on the deviation from the mean pollutant concentration that can be expected in any individual realization. The level of deviation from the corresponding homogeneized (room-averaged) and ensemble-averaged value is measured by the standard deviation $\langle\langle\sigma\rangle\rangle_{\mathrm{dsm}}$. Figure 4 shows the normalized standard deviation, $\langle\langle\sigma\rangle\rangle_{\gamma} = \langle\langle\sigma\rangle\rangle_{\mathrm{dsm}}/\langle\langle n\rangle\rangle_{\mathrm{wmm}}$, plotted as a function of

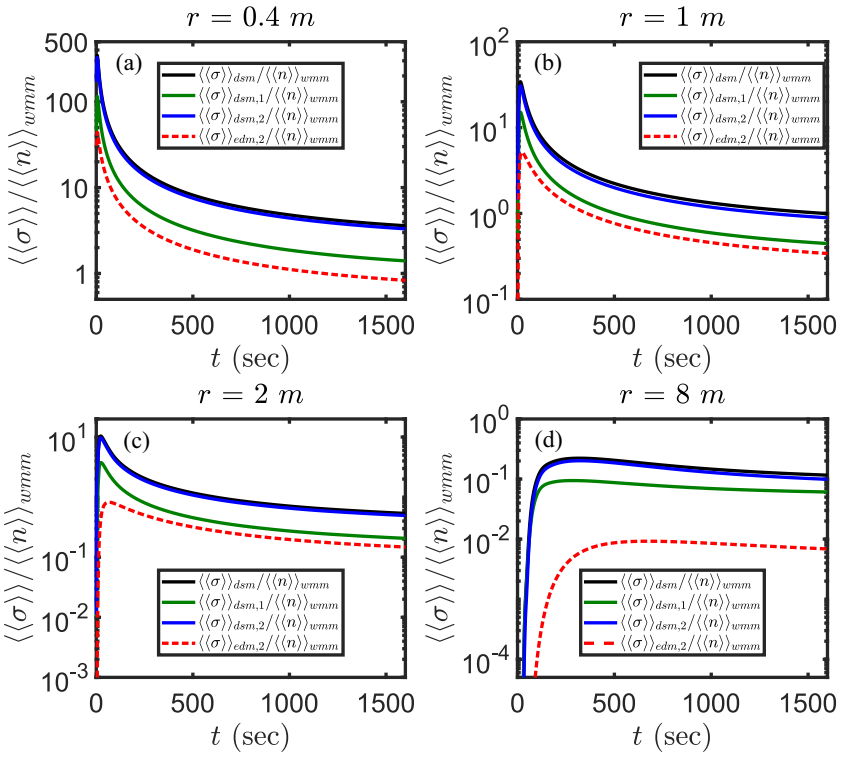


FIG. 4. Comparisons of the temporal evolution of the normalized standard deviations by the models. Plotted here are the total standard deviation predicted by the DSM along with the individual normalized standard deviations due to inhomogeneity and turbulence. Also plotted is the normalized standard deviation due to inhomogeneity predicted by the EDM. Normalized standard deviations are plotted for (a) $r = 0.4$ m, (b) $r = 1$ m, (c) $r = 2$ m, and (d) $r = 8$ m.

t for several values of r. These results were obtained from the ACH = 2.5 LES. Also plotted in this figure are $\langle\langle\sigma\rangle\rangle_{1,\mathrm{dsm}}/\langle\langle n\rangle\rangle_{\mathrm{wmm}}$ and $\langle\langle\sigma\rangle\rangle_{2,\mathrm{dsm}}/\langle\langle n\rangle\rangle_{\mathrm{wmm}}$ evaluated from the simulation. We recall $\langle\langle\sigma\rangle\rangle_{1,\mathrm{dsm}}$ captures the concentration variation due to turbulent fluctuations, while $\langle\langle\sigma\rangle\rangle_{2,\mathrm{dsm}}$ captures the spatial variation due to inhomogeneity within the room. They can be compared with $\langle\langle\sigma\rangle\rangle_{\mathrm{dsm}}/\langle\langle n\rangle\rangle_{\mathrm{wmm}}$, which captures the variation due to both turbulent fluctuations and spatial inhomogeneity within the room. In all cases, the standard deviation rapidly increases, reaches a peak and slowly decays. The peak value is large at smaller distances due to normalization by the well-mixed concentration. As can be seen in Fig. 3, in this normalization, even the mean $\langle\langle n\rangle\rangle_{\mathrm{dsm}}$ is substantially larger than unity at shorter distances.

Normalization by $\langle\langle n\rangle\rangle_{wmm}$ is quite useful since it emphasizes the fact that uncertainty over the well-mixed prediction can be substantial. For example, at $r = 1$ m, according to Fig. 3, the mean concentration can be observed to be about twice that of the WMM prediction and the corresponding standard deviation can be just as large in the first few minutes. Thus, at short times, the measured concentration in a realization can be many times that of the WMM prediction. The total normalized standard deviation, $\langle\langle\sigma\rangle\rangle_{\mathrm{dsm}}/\langle\langle n\rangle\rangle_{\mathrm{wmm}}$ at distances $r = 1$, 2, and 8 m reaches peak values of 1.15, 0.344, and 0.0074 at 15, 23, and 317 s, respectively. Most of the concentration variation is primarily due to spatial variation within the room. Variation due to turbulence is not negligible, although its contribution is smaller.

Among the theoretical models, only the EDM with wall reflections accounts for the specific source and sink locations and therefore is capable of predicting the standard deviation due to spatial

inhomogeneity. Also plotted in Fig. 4 is $\langle\langle\sigma\rangle\rangle_{2,\mathrm{edm}}/\langle\langle n\rangle\rangle_{\mathrm{wmm}}$ for the different distances (red dashed lines). It is quite impressive that the method of reflection is able to account for many of the physics. The rapid rise followed by the slow decay, along with the peak are reasonably well captured by EDM. Quantitatively, the estimated standard deviations are smaller in magnitude and the peaks are shifted to slightly later times. Nevertheless, it appears that the method of reflection can be used to gain some understanding of spatial inhomogeneity in indoor spaces.

B. Dependence on room size, A/C location, and ACH

The dependence of WMM, NFM, and EDM predictions of pollutant concentration on room ventilation, measured in terms of ACH, appears through the removal rate $\lambda(R)$, the random air speed s, and the eddy diffusivity K. In the NFM and EDM, the random air speed s and the eddy diffusivity K can be taken to linearly scale with ACH. The same scaling has been demonstrated for the DSM in the Supporting Information. In this section, we highlight the importance of accounting for statistical variation in several other room and ventilation configurations. These additional results illustrate the broader applicability of the present data-driven results $\langle\langle\gamma\rangle\rangle$ and $\langle\langle\sigma\rangle\rangle_\gamma$. Further generalization for varying geometries and ventilation arrangements is needed to obtain a comprehensive stochastic model.

Choudhary *et al.* [34] performed statistically overloaded RANS simulations of pollutant dispersal in indoor spaces for different room geometries and ventilation systems. Here we utilize results from 3 of their simulations in addition to the canonical room (room 1). Their schematics are shown in Fig. 5. Room 2 is a $7.6 \times 2.3 \times 11.8$ m^3 room with a four-way cassette air conditioning unit similar to the one used in room 1. The AC unit is however shifted by 1.9 m from the center along the longer horizontal dimension. Rooms 3 and 4 have the same dimensions as the canonical room but have differences in the choices of ventilation unit. Room 3 has a wall-mounted one-way AC unit. Room 4 has the same ventilation unit as room 1 but has an open window in addition. More detailed descriptions of the room and ventilation system configurations can be found in Refs. [34,49].

The results for the different rooms showed considerable variation in the behavior of $\langle\langle\gamma\rangle\rangle(t \to \infty, r)$ [34]. Figure 6 compares $\langle\langle\sigma\rangle\rangle_\gamma/\langle\langle\gamma\rangle\rangle$, $\langle\langle\sigma\rangle\rangle_{2,\gamma}/\langle\langle\gamma\rangle\rangle$ and $\langle\langle\sigma\rangle\rangle_{1,\gamma}/\langle\langle\gamma\rangle\rangle$ for the different rooms. To properly scale the different different rooms, time was normalized as $t \times \mathrm{ACH}/3600$ and source-sink distance was normalized by the characteristic dimension of the room as $\tilde{r}$, where the length scale of the room was chosen to be the corner-to-corner longest diagonal within the room. Thus, $\tilde{r} = r/\sqrt{l_x^2 + l_y^2 + l_z^2}$ for a room for dimensions $l_x \times l_y \times l_z$. Figure 6 plots comparisons for the different rooms at $\tilde{r} =$ 0.069 and 0.414 which correspond to $r =$ 1 and 6 m for the canonical room. We note that $\langle\langle\sigma\rangle\rangle_{2,\gamma}/\langle\langle\gamma\rangle\rangle$ shows noticeable variation with room geometry and ventilation system. However, changes in $\langle\langle\sigma\rangle\rangle_{1,\gamma}/\langle\langle\gamma\rangle\rangle$ are minimal for both the values of $\tilde{r}$. In addition to the current comparisons presented for the DSM in the context of varying ventilation flow rates, the effect of the presence of heat sources is discussed in the SM [42] (including Ref. [50]).

V. EXPERIMENTAL VALIDATION

We examine the results of two experimental studies [29,30] to analyze the effectiveness and accuracy of EDM and NFM along with the present data-driven stochastic model. These studies have been chosen for comparison because of the following reasons:

(i) They have been performed over a wide range of ACH values and concentrations have been obtained at multiple separation distances r.

(ii) Concentrations have been measured over multiple sink locations for the same value of r, which allows us to examine the homogenized predictions of the models.

Comparisons with the experiments of Shao *et al.* [29] and Cheng *et al.* [30] are presented in this section.

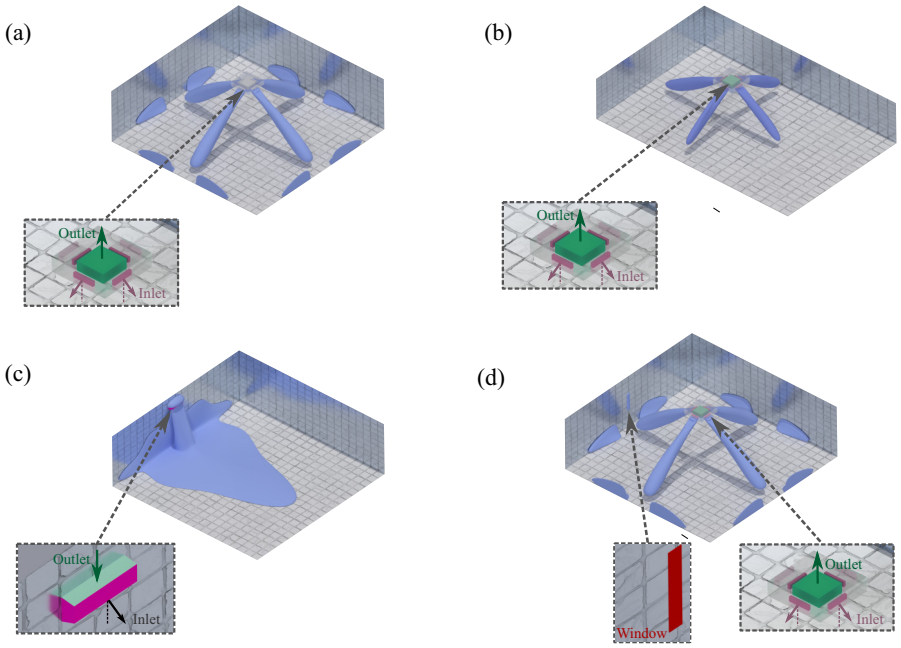


FIG. 5. Layouts of the different geometries for which the simulations were performed along with the velocity magnitude isocontours of 0.6 m/s at ACH of 5 for rooms 1, 2, 3, and 4. LES simulations were performed for room 1, which has been referred to as the canonical room in the main text. RANS simulations were performed for rooms 2, 3, and 4. (a) Room 1: 10 m × 10 m × 3.2 m room with a centrally located four-way cassette air conditioning unit. (b) Room 2: 11.8 m × 7.6 m × 2.3 m room with a four-way cassette air conditioning unit that is shifted 1.9 m from the center of the ceiling along the longer horizontal dimension and is central along the shorter horizontal dimension. (c) Room 4: 10 m × 10 m × 3.2 m room with a wall-mounted one-way cassette air conditioning unit. (d) Room 5: 10 m × 10 m × 3.2 m room with a centrally located four-way cassette air conditioning unit and an open window.

A. Experimental validation—Shao *et al.*

Shao *et al.* [29] performed 27 experiments in an exposure chamber (11.9 m^3, L/W/H: 2.8 m × 2.15 m × 2.0 m) for ACH values ranging from 0.43 to 2.89 with either toluene or acetone as the tracer. The chamber was equipped with four diffusers on one side and zigzagging perforated ducts on the floor to reduce the directionality of airflow and promote turbulent mixing. The point source of the pollutant was placed 0.61 m away from the diffusers along the centerline of the chamber. The gas detectors (sinks) were placed downstream along the centerline of the chamber at a separation distance of 0.41 m (location 1) and 1.07 m (location 2) from the source. Measurements were made over a $T = 3$ h period.

Figures 7(a) and 7(b) provide a comparison of EDM (with reflection terms), NFM, and DSM by plotting their predictions against the experimental measurement of time-averaged concentration. The black solid 45° line indicates the prediction of a perfect model. Figures 7(c)–7(f) provide a comparison of the temporal evolution of concentration for the toluene experiments at locations 1 and 2, respectively, for ACH values of 0.88, 1.31, 2, and 2.62. The model parameters were chosen as described before. The random air speed, s, for the NFM is chosen as 0.06 m/s, as per the guidelines [41]. The value of eddy diffusivity, K, was obtained following the empirical relation provided by Foat *et al.* [48] as it offered higher accuracy than Ref. [45].

Figure 7 illustrates that NFM underpredicts at both locations, with regression line slopes of 0.896 and 0.829, respectively. NFM mean relative errors are 13% and 9.8% at locations 1 and

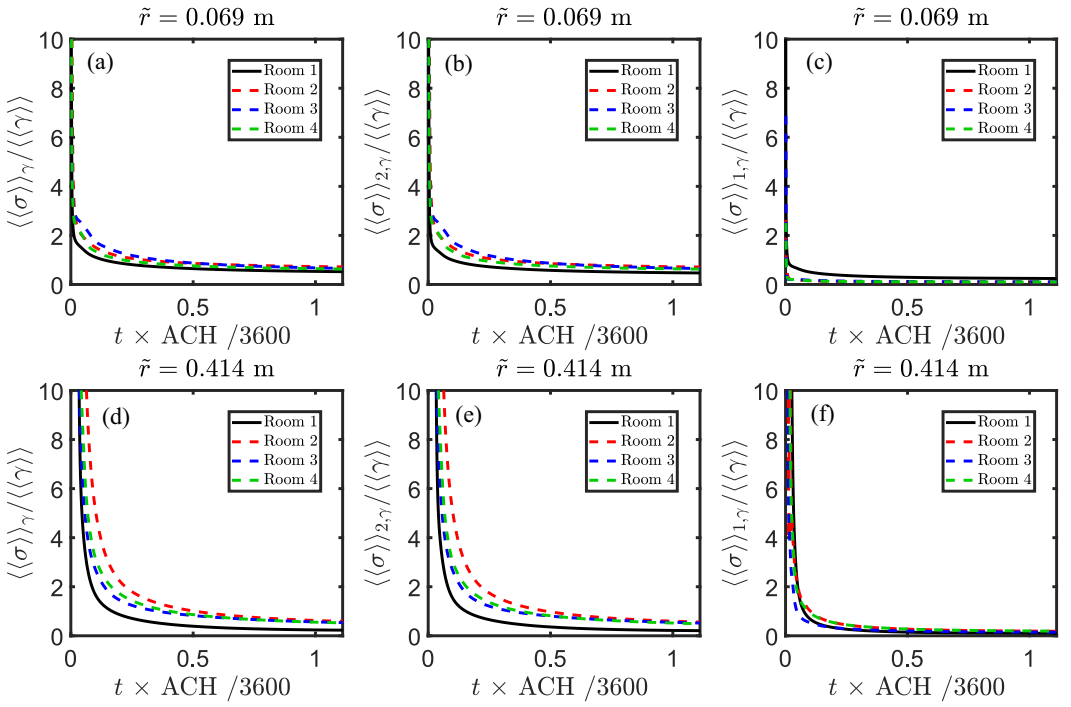


FIG. 6. Comparison of temporal evolution of (a), (d) $\langle\langle\sigma\rangle\rangle_\gamma/\langle\langle\gamma\rangle\rangle$, (b), (e) $\langle\langle\sigma\rangle\rangle_{2,\gamma}/\langle\langle\gamma\rangle\rangle$, and (c), (f) $\langle\langle\sigma\rangle\rangle_{1,\gamma}/\langle\langle\gamma\rangle\rangle$ for $r = 1$ and 6 m for the simulation data from LES and RANS for different room geometries and ventilation systems (rooms 1–4).

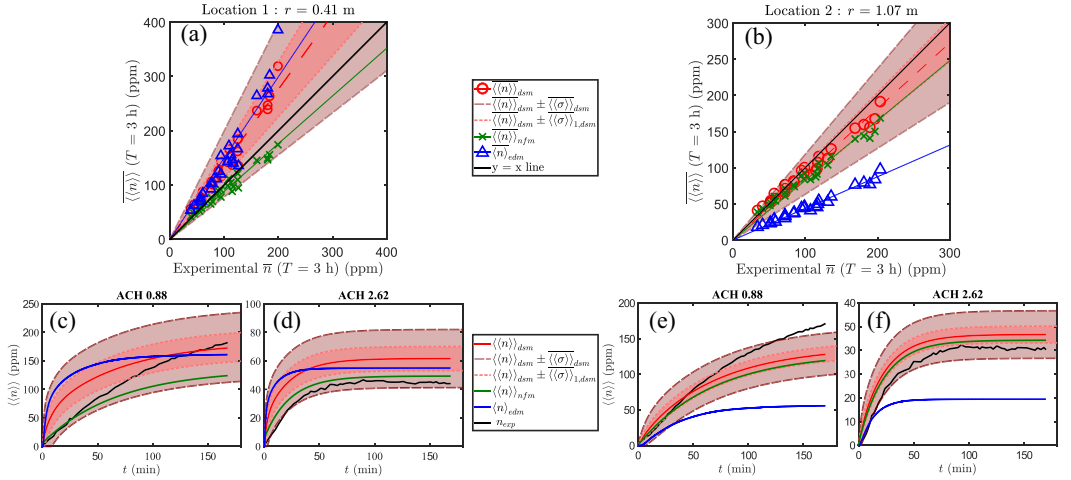


FIG. 7. DSM, NFM, and EDM predictions against experimental data at (a) location 1 ($r = 0.41$ m) and (b) location 2 ($r = 1.07$ m). The lines represent the best fit linear regression for each of these models' predictions. Also plotted are the best-fit linear regressions of the $1-\sigma$ variations of the DSM for $\langle\langle\sigma\rangle\rangle_1$ and $\langle\langle\sigma\rangle\rangle$ with the shaded regions indicating the region of uncertainty due to turbulent fluctuations (darker shade) and total variations (lighter shade), respectively. The rest of the figure comprises of plots providing comparisons of the DSM, NFM, and EDM for the temporal evolution of the concentration observed in the toluene experiments for ACH values of (c), (e) 0.88 and (d), (f) 2.62 for (c), (d) location 1 ($r = 0.41$ m) and (e), (f) location 2 ($r = 1.07$ m).

2, respectively. The parameter s was systematically varied within the range 0.03 to 0.11 m/s but the impact on mean relative error was minimal. The EDM overpredicts at location 1 (Slope: 1.496) and underpredicts at location 2 (Slope: 0.438). The mean relative errors are also significantly higher with values of 39.9% and 55.3% at locations 1 and 2, respectively. These observations are in agreement with those presented in Shao *et al.* [29]. However, the errors they quote are significantly lower due to the fact that they choose the parameter, K, to best fit their data. So models with off-the-shelf parameters may perform poorly, but the performance can be improved by fitting the data.

The stochastic model follows a similar trend as EDM wherein it overpredicts the time-averaged concentration at the closer location (Slope: 1.38) while it underpredicts it at the farther location (Slope: 0.909). However, the errors are lower with mean relative errors of 35.9% for location 1 and 8.72% for location 2. Similar observations can be gleaned from the temporal comparisons where we observe NFM to perform well when compared with the experimental observations. The stochastic model demonstrates effective performance at both locations, although there is a slight overprediction of the mean magnitude.

B. Role of variability

We now examine the nature of the comparison made in Fig. 7. While all three model predictions are for homogenized concentration that is averaged over all possible source-sink locations of specified separation distance, the experimental results are for one particular source-sink combination. When viewed this way, disagreement between the mean prediction by the models and a particular experimental measurement (an individual realization) can be explained as part of the statistical variation. The advantage of DSM is that it can predict the standard deviation of concentration. Figure 7 shows the predicted range of concentration within 1-σ of the mean, represented by the shaded regions, whereby the experimental observations fall well within the 1-σ band. This implies that if the sink had been placed at a different location in the experiment, while maintaining the same separation distance r, then the measured pollutant concentration would likely differ. It must be stressed that if we assume Gaussian statistics, then 32% of the realizations may fall outside the 1-σ band. Nevertheless, the level of stochastic variability in any individual measurement can be estimated with the stochastic model.

C. Experimental validation—Cheng *et al.*

In this subsection, we provide a second experimental comparison by using the experimental results of Cheng *et al.* [30] to further illustrate the arguments made in the previous sections. Cheng *et al.* [30] measured CO concentration due to a continuously emitting point source in two room settings and presented the time-averaged concentration over a $T = 30$ min period. The rooms were naturally ventilated and the corresponding ACH values were measured by releasing SF_6 tracer and then tracking the decay rate. The point source of CO was approximately located at the center of the rooms. Multiple detectors (sinks) were placed radially around the source to measure the concentration at 15 s intervals. 37 monitors were used in room 1 (158 m^3, L/W: 9.4 m $\times$ 4.1 m) while 30 were used for room 2 (59 m^3, L/W: 5.6 m $\times$ 4.4 m). In each room, four monitors were positioned radially at separation distances of $d = 0.25$, 0.5, 1, and 2 m. Additionally, room 1 had two monitors at 3 and 4 m, and one at 5 m. For room 2, two monitors were placed at 2.8 m and four were placed in the corners of the room (3.56 m). Cheng *et al.* [30] performed five experiments in room 1 and six experiments in room 2 for ACH values ranging from 0.17 to 5.4.

Figure 8 shows the comparison of the predictions of the different models with the measured experimental results. The time-averaged concentration has been normalized by the WMM prediction and plotted on the y axis as a function of r. All 11 experiments were compared, but only 4 are presented here as they concisely summarize the conclusions. In the NFM, the random airspeed is set at $s = 0.06$ m/s, in accordance with Nicas' suggestion [51]. Given that the rooms were naturally

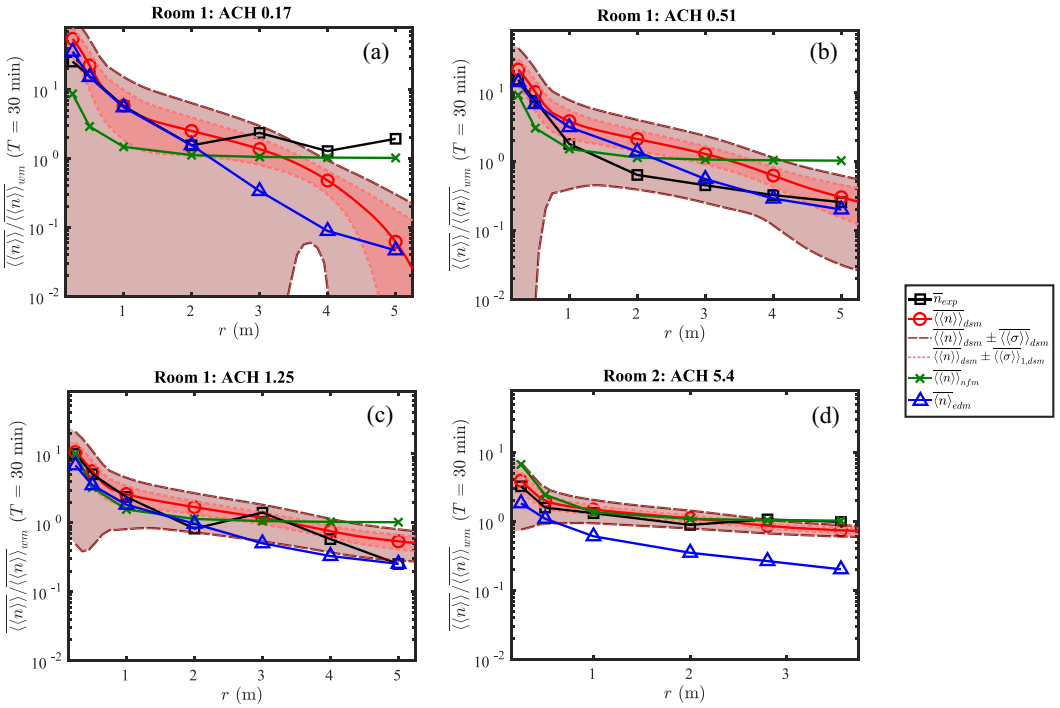


FIG. 8. The time-averaged concentrations normalized by the time-averaged concentration predicted by the WMM are plotted as a function of separation distance for the experimental results of Cheng *et al.* [30] along with the predictions of the stochastic, eddy diffusivity, and the near-field/far-field models for ACH values of 0.17, 0.51, 1.25, 5.4.

ventilated, we applied Venkatram and Weil's [45] relationship for K in the EDM. We considered both EDM with and without the influence of reflections.

The NFM performs well at the higher ACH values compared to the lower ACH values. Even at the higher ACH, NFM prediction is higher at the lower separation distances. The eddy diffusivity model, however, performs better at the lower values of ACH (0.17, 0.51). Since one of the assumptions of the EDM is that the flow lacks an advective component, it justifies its better performance at lower ACH values where the advective effect is weaker.

The stochastic model seems to perform relatively well at all values of ACH. However, at smaller values of ACH, it produces a significantly higher error at large separation distances. This can be explained by the fact that the stochastic model was developed using LES data from forced ventilation with values of ACH larger than unity. Due to the substantially larger time scales associated with ACH values of 0.17 and 0.51 compared to those used in the simulations, the stochastic model underestimates the dominance of diffusion over advective transport. This leads to a lower prediction of pollutant concentration.

As discussed previously, one of the advantages of the stochastic model is its ability to predict the standard deviation in addition to the mean value. 1-σ deviation band around the mean is shown in Fig. 8 and the results of both variation due to only turbulence ($\langle\langle\sigma\rangle\rangle_{1,\mathrm{ds}}$) and total variation ($\langle\langle\sigma\rangle\rangle_{\mathrm{ds}}$) are shown. Except at large distance values, the experimental results are well captured by the 1-σ variation. In the experiments, there were fewer sensors at larger separation distances, and as a result, the experimental result better approximates an individual realization than an ensemble average that has been spatially homogenized. This is confirmed by the fact that the time-averaged concentration at larger separation distances are larger than the room-averaged prediction of the WMM, whereas, we expect the concentration at larger distances to be lower than the room average

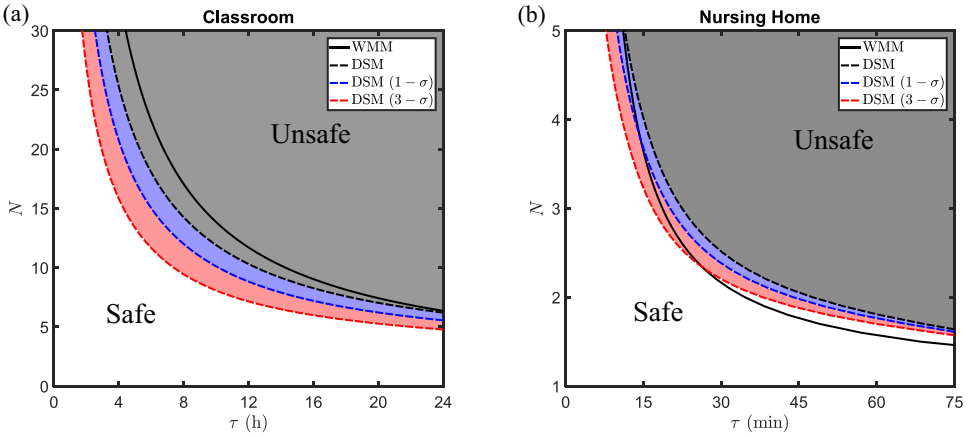


FIG. 9. Indoor safety guideline for maximum safe occupancy time against occupancy limit devised based on Bazant and Bush's study [38] for a typical (a) classroom and (b) nursing home. The solid black and red lines represent the threshold safety limit for the WMM and mean prediction of the DSM, respectively, while the different dashed lines represent the 1-σ and 3-σ variations above the mean. For (a) the classroom with a floor area of 83.6 m^2 and volume of 301 m^3, a low relative transmissibility ($s_r = 0.1$) and moderate risk tolerance ($\epsilon = 0.1$) are chosen. For (b) the nursing home with a floor area of 22.3 m^2 and volume of 53.5 m^3, a high relative transmissibility ($s_r = 1$) and low risk tolerance ($\epsilon = 0.01$) are chosen. Both cases assume masks are not in use ($p_m = 1$). Other parameters are ACH = 8, $Q_b = 0.5$ m^3/h and $C_q = 30$ quanta/m^3.

to compensate for the higher-than-average concentration in the near-field. However, this was not the case in the experimental measurement, perhaps due to systematic bias induced by the location of the measurement probes or due to random turbulent fluctuations. Irrespective, only with proper estimation of standard deviation one can account for such deviations from the predicted mean.

D. Impact of uncertainty on safety guideline

The DSM can be easily implemented to obtain improved safety guidelines for different indoor environments by fully accounting for the uncertainties expected in individual realizations. This capability is showcased in Fig. 9 for the two cases of a classroom and a nursing home considered by Bazant and Bush [38]. Their results of cumulative exposure time (CET) using the WMM for a ventilation system with ACH = 8 and a perfect filtration system are shown as solid black lines. Figures 9(a) and 9(b) showcase the guidelines for a typical school classroom and a nursing home. The CET plots display the occupancy, N, i.e., the number of people simultaneously occupying the indoor space against the maximum occupancy time, τ, that the occupants can share the space without reaching the risk threshold. The results of DSM obtained with the optimal distance between the occupants chosen to be $r = \sqrt{A/N}$ is shown as the black dashed line. Also plotted are the 1-σ and 3-σ predictions of the DSM above the mean (see SM [42] for additional details). The N-τ combinations below the lines depict the safe combinations of occupancy and exposure times.

The impact of including uncertainty can now be assessed. For the classroom, for a nominal occupancy of 20, based on average viral nuclei concentration the WMM and DSM predict safe occupancy times of 6.8 and 5.3 h, respectively. If we consider the worst-case scenario of the negative 3-σ curve (corresponding to 99.7% confidence interval), then the occupancy time drops to 2.95 h. Similarly, for a 10 min meeting inside a nursing home, the mean prediction of the DSM yields a maximum occupancy of five while the 3-σ variation allows only a maximum occupancy of three. The significant changes in safety guidelines indicate the importance of accounting for the variability expected in individual realizations.

VI. CONCLUSIONS

We analyzed results from turbulence-resolving, Euler-Lagrange simulations of pollutant dispersal within different indoor settings for 25 million source-sink combinations to assess the uncertainty in the prediction of the mean pollutant concentration. The uncertainty primarily arises from two sources:

(i) Turbulent fluctuations that are inherent to the chaotic nature of the flow, which We quantify by considering an ensemble of realizations.

(ii) Variability (or inhomogeneity) within the room due to the presence of walls and proximity to inlet jets. In general, pollutant concentration at a sink depends on the specific locations of both the sink and the source within the room.

Dispersal models employ the assumption of spatial homogenization, where only the separation distance is considered. However, in a real application, it matters whether the sink is located directly downstream or upstream of the source in the path of air circulation. Here we exploit the statistical overloading technique to quantify the uncertainty due to homogenization by considering all possible source-sink combinations for a particular separation distance. We observe that in advection-dominated regions of the indoor space, the uncertainty due to inhomogeneity is substantial compared to regions where pollutants are primarily transported by turbulent diffusion.

The simulation results were used to propose a DSM for the prediction of pollutant concentration. The model is compared to experimental data as well as other models on pollutant dispersal, including the WMM, the NFM, and the EDM. The data-driven stochastic model is built upon the WMM by proposing a correction function that accurately accounts for the concentration dependence on source-sink separation distance. Most importantly, DSM extends beyond the mean concentration prediction, by providing a precise estimate of the standard deviation. This estimate of standard deviation accounts for both uncertainty due to turbulence and inhomogeneity resulting from spatial variation.

We observe that the NFM performs well at higher ACH values of $\mathcal{O}(>1)$ compared to lower ACH values. It correctly captures the rapid short-time advective increase followed by the long-time slower dispersive increase in concentration for various source-to-sink separation distances. Its minor drawback is that it consistently overpredicts the concentration at larger separation distances. However, the EDM performs well at lower ACH values and at shorter separation distances, since one of its assumptions is that the flow lacks an advective component. EDM is observed to considerably underpredict pollutant concentration at larger separation distances despite accounting for the effect of reflection.

It should be emphasized that in an experiment, the time-averaged concentration measurement made at a single source-sink combination represents an individual realization. Whereas, the models attempt to predict the ensemble-averaged cumulative concentration over all possible source-sink combinations. Thus, a perfect match between the experiments and the models should not be expected. The present stochastic model predicts not only the mean but also the standard deviation and has been shown to accurately estimate the band within which the cumulative concentration of an individual realization will fall. Such an ability to accurately predict both the expected exposure and the associated uncertainty has been demonstrated to significantly alter the risk assessment.

Another important consideration to be kept in mind is that the stochastic model was developed based on results from a canonical room and ventilation system configuration. The nature of turbulent dispersal of aerosols in indoor flows is heavily dictated by the directionality of flow patterns. As highlighted in the Supporting Information, changes in the room and ventilation system will affect the stochastic model's performance. In addition to this, turbulent dispersal in larger spaces like auditoriums or restaurants and transportation systems like trains or airplanes might necessitate the use of a multizonal model involving exchange between the different individual well-mixed zones

[52,53]. The applicability of the stochastic model in such scenarios is also something that needs to be explored and validated.

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K.A.K. conducted the simulations and processed the data. S.B., N.Z., and K.A.K. conceptualized the work. S.B., N.Z., and K.A.K. wrote the paper.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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